

# Artificial Intelligence Overview

Artificial Intelligence (AI) is transforming industries by enabling machines to perform tasks that traditionally required human intelligence. Machine Learning is a subset of AI that learns patterns from data. Supervised learning uses labeled data, while unsupervised learning discovers hidden structures. Deep Learning uses neural networks with many layers and powers applications such as image recognition, natural language processing, and autonomous systems. Modern AI systems are used in healthcare, finance, education, and transportation. In healthcare, AI assists doctors in diagnosis. In finance, AI helps detect fraud and automate trading decisions. Large Language Models (LLMs) are a recent advancement that can understand and generate human-like text. They are trained on large datasets and can assist with coding, writing, and research tasks. Challenges in AI include data privacy, bias, explainability, and energy consumption. Responsible AI development focuses on fairness, transparency, and accountability. Future developments may include more capable multimodal systems that understand text, images, audio, and video simultaneously.

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